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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
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EXAMINER	
GAY, JENNIFER HAWKINS	

ART UNIT	PAPER NUMBER
3676	

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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Office Action Summary	Application No.	Applicant(s)	
	12/238,086	LUMBYE, PETER	
	Examiner	Art Unit	
	JENNIFER H. GAY	3676	

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE *3* MONTH(S) OR THIRTY (30) DAYS, WHICHEVER IS LONGER, FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☐ Responsive to communication(s) filed on _____.

2a)☐ This action is **FINAL**. 2b)☒ This action is non-final.

3)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims

4)☒ Claim(s) *1-16* is/are pending in the application.

4a) Of the above claim(s) _____ is/are withdrawn from consideration.

5)☐ Claim(s) _____ is/are allowed.

6)☒ Claim(s) *1-16* is/are rejected.

7)☐ Claim(s) _____ is/are objected to.

8)☐ Claim(s) _____ are subject to restriction and/or election requirement.

Application Papers

9)☐ The specification is objected to by the Examiner.

10)☒ The drawing(s) filed on *04 December 2008* is/are: a)☐ accepted or b)☒ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

11)☐ The oath or declaration is objected to by the Examiner. Note the attached Office Action or form PTO-152.

Priority under 35 U.S.C. § 119

12)☒ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

a)☒ All b)☐ Some * c)☐ None of:

1.☒ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

* See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☒ Notice of References Cited (PTO-892)
2) ☐ Notice of Draftsperson's Patent Drawing Review (PTO-948)
3) ☒ Information Disclosure Statement(s) (PTO/SB/08) Paper No(s)/Mail Date _____.
4) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date. _____ .
5) ☐ Notice of Informal Patent Application
6) ☐ Other: _____.

DETAILED ACTION

Drawings

1. The drawings are objected to under 37 CFR 1.83(a). The drawings must show every feature of the invention specified in the claims. Therefore, the **distribution of the perforations along the tubular or casing as recited in claims 7-12, 14, and 15** must be shown or the feature(s) canceled from the claim(s). No new matter should be entered.

Corrected drawing sheets in compliance with 37 CFR 1.121(d) are required in reply to the Office action to avoid abandonment of the application. Any amended replacement drawing sheet should include all of the figures appearing on the immediate prior version of the sheet, even if only one figure is being amended. The figure or figure number of an amended drawing should not be labeled as "amended." If a drawing figure is to be canceled, the appropriate figure must be removed from the replacement sheet, and where necessary, the remaining figures must be renumbered and appropriate changes made to the brief description of the several views of the drawings for consistency. Additional replacement sheets may be necessary to show the renumbering of the remaining figures. Each drawing sheet submitted after the filing date of an application must be labeled in the top margin as either "Replacement Sheet" or "New Sheet" pursuant to 37 CFR 1.121(d). If the changes are not accepted by the examiner, the applicant will be notified and informed of any required corrective action in the next Office action. The objection to the drawings will not be held in abeyance.

Claim Objections

2. Claim **3** is objected to because of the following informalities: **"establishment" should be changed to --the establishment--**. Appropriate correction is required.

Claim Rejections - 35 USC § 112

3. The following is a quotation of the second paragraph of 35 U.S.C. 112:

> The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

4. **Claims 1-16 are rejected under 35 U.S.C. 112, second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which applicant regards as the invention.**

Regarding claim 1: Lines 9-11 are considered to be an improper Markush listing. Alternate expressions are permitted if they present no uncertainty or ambiguity with respect to the question of scope or clarity of the claims. On acceptable form of alternative expression, or Markush group, recites members as being "selected from the group consisting of A, B, or C." See Ex parte Markush, 1925 C.D. 126 (Comm'r5 Pat. 1925). (See MPEP 2173.05(h))

Regarding claim 5: Lines 8-11 are considered to be an improper Markush listing. Alternate expressions are permitted if they present no uncertainty or ambiguity with respect to the question of scope or clarity of the claims. On acceptable form of alternative expression, or Markush group, recites members as being "selected from the group consisting of A, B, or C." See Ex parte Markush, 1925 C.D. 126 (Comm'r5 Pat. 1925). (See MPEP 2173.05(h))

Regarding claim 13: The preamble of the claim recites "A tubular used for…" however the body of the claim includes such features as "well casing", "tubing", and "packers" all of which cannot be considered part of a "tubular". For the purposes of examination, the examiner is treating claim 13 as reciting --A system for--.

Regarding claims 2-4, 6-12, and 14-16: These claims are rejected as being dependent upon indefinite claims.

Claim Rejections - 35 USC § 102

5. The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

A person shall be entitled to a patent unless –

(b) the invention was patented or described in a printed publication in this or a foreign country or in public use or on sale in this country, more than one year prior to the date of application for patent in the United States.

(e) the invention was described in (1) an application for patent, published under section 122(b), by another filed in the United States before the invention by the applicant for patent or (2) a patent granted on an application for patent by another filed in the United States before the invention by the applicant for patent, except that an international application filed under the treaty defined in section 351(a) shall have the effects for purposes of this subsection of an application filed in the United States only if the international application designated the United States and was published under Article 21(2) of such treaty in the English language.

6. **Claims 1-4 and 7-12 are rejected under 35 U.S.C. 102(e) as being anticipated by Matthew (US 7,401,652).**

Regarding claim 1: Matthew discloses a method for stimulating a well for production of hydrocarbons from a formation when a stimulation fluid is introduced through a cemented and perforated tubular for dissolving material **(It is noted that while the overall purpose of the method of Matthew is to fracture the wellbore, Matthew does disclose using acid prior to the fracturing fluid being pumped in the wellbore.)**, the method involves the following the steps:

- establishing a wellbore;
- introducing a tubular into the wellbore for the production of hydrocarbons;
- cementing the tubular for retaining the tubular in the wellbore **5:45-59**;
- pre-selecting locations to perforate the tubular in the wellbore by modeling inflow performance of the stimulation fluid in the formation **(Matthew discloses testing and measuring fracture characteristics such as entry pressure which implies measuring the ability of a stimulation fluid to flow into the formation)**, based on parameters of the composition of the formation, the parameters including reservoir porosity, permeability, pressure, damage skin and intrusion depth, perforation diameter and penetration depth or perforation crushed zone damage and intrusion depth **6:16-30 and 6:61-7:3**;
- perforating the tubular at the pre-selected locations after the tubular has been cemented in place **7:4-21**; and

- stimulating the well by introducing the stimulation fluid into the formation through the perforated tubular at the pre-selected locations **8:4-15**.

Regarding claim 2: The pre-selected locations where the tubular is perforated are determined by calculating an inflow performance individually for each perforation while the inflow performance changes as a function of inflow volume of the stimulation fluid **6:16-30 and 6:61-7:3**.

Regarding claim 3: The method involves logging data regarding the formation and reservoir prior to the fracturing operation where the logging is performed using conventional methods **5:60-6:15**. While it is not specifically disclosed that the logging is performed during the "establishment" of the wellbore, logging while drilling is a known and conventional means by which a wellbore is logged.

Regarding claim 4: The method involves calculating a stimulation treatment based on variations of one or more parameters of reservoir properties **6:61-7:3**.

Regarding claims 7-12: Matthews discloses choosing a perforation pattern to optimize the fracture performance. This involves different spacing and different densities of perforations in clusters **(7:4-8:3)**.

7. **Claims 13 and 16 are rejected under 35 U.S.C. 102(b) as being anticipated by Kilgore (US 6,547,011).**

Regarding claim 13: Kilgore discloses a system **Figure 2** for use in a wellbore for the production of hydrocarbons; the system includes the following features:

- a well casing **(12, 4:1-17)** with perforations **(the casing has to include perforations in order for fluid to be produced therethrough)** along

the length of the well casing, the perforations permitting fluid flow between the interior and exterior of the well casing;

- a tubing **28** inside the well casing comprising closable openings **36, 38, 40 (5:14-29)** that regulate fluid flow from the interior of the tubing to a zone in an annular space exterior to the tubing but interior to the well casing; and
- a plurality of packers **30, 32, 34** spacing the tubing from the well casing and forming the zone.

Regarding claim 16: The closable openings are closed in the proximity of perforations in the well casing where fluid flow is not desired and the closable openings are open in the proximity of perforations in the well casing where fluid flow is desired **5:14-29**.

Claim Rejections - 35 USC § 103

8. The following is a quotation of 35 U.S.C. 103(a) which forms the basis for all obviousness rejections set forth in this Office action:

> (a) A patent may not be obtained though the invention is not identically disclosed or described as set forth in section 102 of this title, if the differences between the subject matter sought to be patented and the prior art are such that the subject matter as a whole would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject matter pertains. Patentability shall not be negatived by the manner in which the invention was made.

9. **Claims 1-4 and 7-12 are rejected under 35 U.S.C. 102(e) as anticipated by (rejection given above) or, in the alternative, under 35 U.S.C. 103(a) as obvious over Matthew in view of WO 2005/064117 (referred to hereafter as Pribnow; US 7,475,724 is the US equivalent and will be referred to in the rejection below).**

Regarding claims 1 and 2: Matthew, as described above, discloses all of the limitations of the above claim(s) except for the pre-selected locations where the tubular is perforated are determined by calculating an inflow performance individually for each perforation while the inflow performance changes as a function of inflow volume of the

stimulation fluid based on modeling inflow performance of the stimulation fluid in the formation.

Pribnow discloses a method of determining the inflow of fluid into a given formation. Pribnow further discloses using a simulation or a model to obtain this data for specific locations in the formation **2:32-53 and 8:27, 28**.

It would have been considered obvious to one of ordinary skill in the art, at the time the invention was made, to have modified the method of Matthew to use a model to determine the inflow performance of the stimulation fluid in the formation at specific locations as suggested by Pribnow in order to have obtained the necessary data without disturbing or possibly damaging the producing formation.

Regarding claim 3: The method of Matthew involves logging data regarding the formation and reservoir prior to the fracturing operation where the logging is performed using conventional methods **5:60-6:15**. While it is not specifically disclosed that the logging is performed during the "establishment" of the wellbore, logging while drilling is a known and conventional means by which a wellbore is logged and thus the examiner takes **OFFICAL NOTICE** that it would have been considered obvious to one of ordinary skill in the art, at the time the invention was made, to have obtained the formation and reservoir data while the wellbore was being established in order to have avoided the need to run separate tools and operations to obtain the needed wellbore logs.

Regarding claim 4: The method of Matthew involves calculating a stimulation treatment based on variations of one or more parameters of reservoir properties **6:61-7:3**.

Regarding claims 7-12: Matthews discloses choosing a perforation pattern to optimize the fracture performance. This involves different spacing and different densities of perforations in clusters **(7:4-8:3)**. Therefore, it would have been considered obvious to one of ordinary skill in the art, at the time the invention was made, to have oriented/placed the perforations in what ever pattern or density would have resulted in optimized production.

10. **Claim 6 is rejected under 35 U.S.C. 103(a) as being unpatentable over Matthew in view of Misselbrook et al. (US 6,497,290, referred to hereafter as Missel).**

Matthew discloses all of the limitations of the above claim(s) except for the stimulation fluid including friction reducers.

Missel discloses a method for stimulating a wellbore that involves fracturing where the fracturing fluid can include friction reducers **6:27-53**.

It would have been considered obvious to one of ordinary skill in the art, at the time the invention was made, to have modified the stimulation fluid of Matthew to include friction reducers as taught by Missel in order to have allowed for easier and more effective proppant placement in the fractures. This would have resulted in the predictable result of a better packed fracture thus enhancing the productivity of that fracture.

11. **Claims 14 and 15 are rejected under 35 U.S.C. 103(a) as being unpatentable over Kilgore in view of Matthew.**

Kilgore discloses all of the limitations of the above claim(s) except for the perforations being distributed in clusters along the length of the well casing where each cluster has a different perforation density.

Matthews discloses choosing a perforation pattern to optimize the fracture performance. This involves different spacing and different densities of perforations in clusters **(7:4-8:3)**.

Therefore, it would have been considered obvious to one of ordinary skill in the art, at the time the invention was made, to have oriented/placed the perforations of Kilgore in what ever pattern or density would have resulted in optimized production as taught by Matthew.

Allowable Subject Matter

12. Claim **5** would be allowable if rewritten to overcome the rejection(s) under 35 U.S.C. 112, 2nd paragraph, set forth in this Office action and to include all of the limitations of the base claim and any intervening claims.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to JENNIFER H. GAY whose telephone number is (571)272-7029. The examiner can normally be reached on Monday through Friday, 8 am to 3 pm.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, David Bagnell can be reached on (571) 272-6999. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see http://pair-direct.uspto.gov. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/Jennifer H Gay/
Primary Examiner, Art Unit 3676

JHG
3/4/10